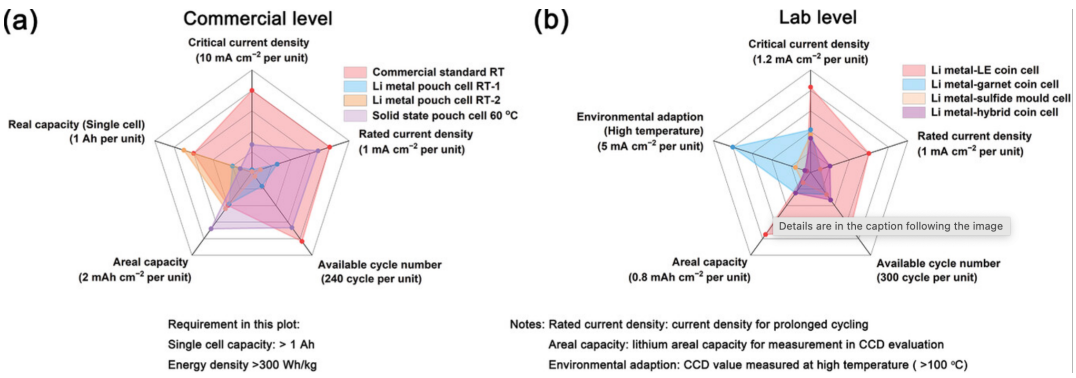
(A) Chemical properties

1) Critical Current Density, CCD

Technical stage	CCD range	Influence
Liquid electrolyte	0.5–1 mA/cm²	Dendrite
Advanced liquid electrolyte	1–3 mA/cm²	A part of dendrite free
Solid state electrolyte	1–5 mA/cm²	Unstable interface
Advanced SSE + interface energy	5–10 mA/cm²	Approaching practicality
2035+	>10 mA/cm²	Fast-charging level



By 2025, higher CCD technology will be achieved and the batteries will be able to charge quickly.

Timeframe	Research priority	Target KPI
2025–2030	Stable SEI design	CCD > 5 mA/cm²
2035–2040	Industrial LMB	CCD > 10 mA/cm²

[1] Lu, Yang, et al. "Critical current density in solid-state lithium metal batteries: mechanism, influences, and strategies." *Advanced Functional Materials* 31.18 (2021): 2009925.

[2] Liu, Gaozhan, et al. "Densified Li6PS5Cl nanorods with high ionic conductivity and improved critical current density for all-solid-state lithium batteries." *Nano letters* 20.9 (2020): 6660-6665.

[3] Fuchs, Till, et al. "Evaluating the use of critical current density tests of symmetric lithium transference cells with solid electrolytes." *Advanced Energy Materials* 13.45 (2023): 2302383.

[4] Gao, Min, et al. "Constructing a multifunctional interlayer toward ultra-high critical current density for garnet-based solid-state lithium batteries." Advanced Functional Materials 33.22 (2023): 2300319

[5] Pang, Bo, et al. "Ag nanoparticles incorporated interlayer enables ultrahigh critical current density for Li6PS5Cl-based all-solid-state lithium batteries." Journal of Power Sources 563 (2023): 232836.

2) Cycle life

cycle life to 80% capacity retention

Technical level	**Cycle life**	**Definition**
Traditional Li mental	<100 cycles	Unstable
Lab system	100–300 cycles	Use in lab
Advanced solid state system	300–1000 cycles	Approaching practicality
2035+	800–1500+ cycles	High level

Timeframe	Research focus	Target KPI
2025–2030	Interface stabilization	>300 cycles
2035–2040	Industrial LMB	>1500 cycles

[1] Yao, Xiayin, et al. "High-energy all-solid-state lithium batteries with ultralong cycle life." *Nano letters* 16.11 (2016): 7148-7154.

[2] Zhang, Zhihua, et al. "An advanced construction strategy of all-solid-state lithium batteries with excellent interfacial compatibility and ultralong cycle life." *Journal of Materials Chemistry A* 5.32 (2017): 16984-16993.

[3] Wu, Feng, et al. "“Liquid-in-solid” and “solid-in-liquid” electrolytes with high rate capacity and long cycling life for lithium-ion batteries." *Chemistry of Materials* 28.3 (2016): 848-856

[4] Choi, S. S., & Lim, H. S. (2002). Factors that affect cycle-life and possible degradation mechanisms of a Li-ion cell based on LiCoO2. Journal of power sources, 111(1), 130-136.

3) Coulombic Efficiency

Technical level	CE	**Definition**
Traditional Li mental	95–98%	Sharp decline
Lab system	98–99.5%	Controllable
Advanced system	99.5–99.8%	Approaching practicality
2035+	≥ 99.9%	High level

Timeframe	**Target CE**
2025–2030	≥99.5%
2035–2040	≥99.9%

[1] Yang, Fangfang, et al. "A study of the relationship between coulombic efficiency

and capacity degradation of commercial lithium-ion batteries." *Energy* 145 (2018): 486-495.

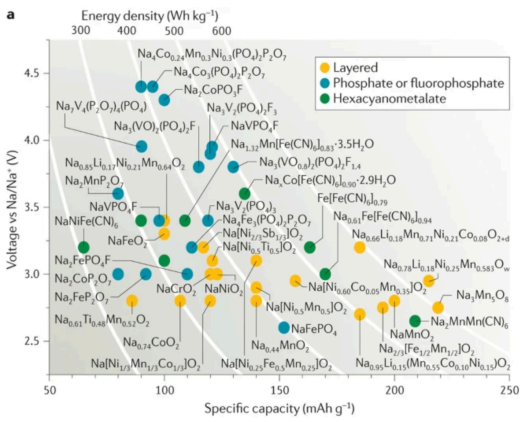
[2] Xiao, Jie, et al. "Understanding and applying coulombic efficiency in lithium metal batteries." *Nature energy* 5.8 (2020): 561-568.

4) Energy density

Type	Energy density
Traditional lithium battery	50–300 Wh/kg
Lab battery	350–450 Wh/kg
2035+	450–500+ Wh/kg

Energy density remains the primary motivation for lithium metal batteries. Near-term development targets aim for 350–400 Wh kg^{-1} at the cell level, while long-term commercialization requires ≥500 Wh kg^{-1} to surpass the performance ceiling of conventional lithium-ion technology.

Timeframe	Energy density
2025–2030	350–400 Wh/kg
2035–2040	≥500 Wh/kg



[1] Li, Jianlin, et al. "Toward low-cost, high-energy density, and high-power density lithium-ion batteries." *Jom* 69.9 (2017): 1484-1496

5) Areal capacity

.Areal capacity is a critical engineering parameter reflecting the practical loading level of electrodes. Near-term research aims to achieve ≥3 mAh cm^{-2}, while commercial lithium metal batteries require ≥5 mAh cm^{-2} to meet industrial energy density targets.

Timeframe	Areal capacity
2025–2030	≥3 mAh/cm²
2035–2040	≥5 mAh/cm²

[1] Gaikwad, Abhinav M., et al. "A high areal capacity flexible lithium-ion battery with a strain-compliant design." *Advanced Energy Materials* 5.3 (2015): 1401389.

[2] Gallagher, Kevin G., et al. "Optimizing areal capacities through understanding the limitations of lithium-ion electrodes." *Journal of The Electrochemical Society* 163.2 (2016): A138-A149.

6) Operating Temperature Window

Operating temperature window is a critical reliability metric governing dendrite kinetics and interfacial stability. Near-term targets aim for stable cycling between 0–50 °C, while long-term industrial deployment requires a wide operating range from −20 °C to 80 °C.

Temperature range	Effect
< 0°C	Rapid dendrite growth
0–25°C	Laboratory standard range
25–45°C	Most stable range
> 60°C	Thermal failure risk

Timeframe	**Temperature**
2025–2030	0–50°C
2035–2040	−20–80°C

[1] Ma, Shuai, et al. "Temperature effect and thermal impact in lithium-ion batteries: A review." *Progress in Natural Science: Materials International* 28.6 (2018): 653-666.

[2] Aris, Asma Mohamad, and Bahman Shabani. "An experimental study of a lithium ion cell operation at low temperature conditions." *Energy Procedia* 110 (2017): 128-135.

[3] Lu, Z., et al. "A comprehensive experimental study on temperature-dependent performance of lithium-ion battery." *Applied Thermal Engineering* 158 (2019): 113800.

(B) Material and interface

1) Mechanical modulus

Mechanical modulus is a key parameter governing dendrite penetration resistance. Based on the Monroe–Newman criterion, electrolytes with shear modulus exceeding 10 GPa are desirable. Near-term targets focus on achieving 5–10 GPa composite electrolytes, while long-term development aims for ≥20 GPa materials with enhanced fracture tolerance.

Liquid

Timeframe	Mechanical modulus
2025–2030	≥5–10 GPa
2035–2040	≥20 GPa

solid

Timeframe	Mechanical modulus
2025–2030	≥200 GPa
2035–2040	≥300 GPa

[1] Wolfenstine, Jeff, et al. "Mechanical behavior of Li-ion-conducting crystalline oxide-based solid electrolytes: a brief review." *Ionics* 24.5 (2018): 1271-1276.

[2] Gucci, Francesco, et al. "Electrical and mechanical characterisation of poly (ethylene) oxide-polysulfone blend for composite structural lithium batteries." *Polymers* 15.11 (2023): 2581.

[3] Tripathi, S. K., Ashish Gupta, and Manju Kumari. "Studies on electrical conductivity and dielectric behaviour of PVdF–HFP–PMMA–NaI polymer blend electrolyte." *Bulletin of Materials Science* 35.6 (2012): 969-975.

[4] Yue, Liping, et al. "All solid-state polymer electrolytes for high-performance lithium ion batteries." *Energy Storage Materials* 5 (2016): 139-164

2) Ionic conductivity

Ionic conductivity governs lithium-ion transport uniformity and is directly linked to dendrite suppression capability. Near-term development targets aim for $\sigma \geq 10^{-3}$ S cm^{-1}, while long-term industrial solid electrolytes require $\sigma \geq 10^{-2}$ S cm^{-1} to match liquid-electrolyte performance.

Electrolyte type	$\sigma_{(ion)}$
Liquid electrolyte	10^{-2} S/cm
Polymer electrolyte	10^{-5} – 10^{-4} S/cm
Composite electrolyte	10^{-4} – 10^{-3} S/cm
Excellent solid sulfide	10^{-3} – 10^{-2} S/cm
Ideal target	$\geq 10^{-2}$ S/cm

Timeframe	Ionic conductivity
2025–2030	$\geq 10^{-3}$ S/cm
2035–2040	$\geq 10^{-2}$ S/cm

These data come from Table 2 & Table 3 in the review paper.

3)Electronic conductivity

Electronic conductivity must remain extremely low to prevent internal lithium plating within the electrolyte. Near-term targets aim for $\sigma_e \leq 10^{-10}$ S cm^{-1}, while long-term industrial electrolytes require $\sigma_e \leq 10^{-12}$ S cm^{-1} to ensure dendrite-free operation.

Timeframe	**Electronic conductivity**
2025–2030	$\leq 10^{-10}$ S/cm
2035–2040	$\leq 10^{-12}$ S/cm

[1] Thangadurai, V., & Weppner, W. (2006). Recent progress in solid oxide and lithium ion conducting electrolytes research. *Ionics*, *12*(1), 81-92.

4）**Defect density**

Including: void fraction (%), crack density (mm^{-2}), grain boundary density, defect per μm^3

Microstructural defect density critically determines dendrite nucleation sites. Near-term targets aim to reduce electrolyte porosity below 3%, while long-term industrial benchmarks require sub-1% porosity and crack-free interfaces to eliminate defect-driven dendrite initiation.

Electrolyte state	Defect level
Conventional sintering SSE	5–10% porosity
Optimized process	<3% porosity
High-end dense SSE	<1% porosity
Ideal target	Approaching 0

Timeframe	**Defect density**
2025–2030	<3% porosity
2035–2040	<1% porosity

[1] Gao, Zhonghui, et al. "Promises, challenges, and recent progress of inorganic solid-state electrolytes for all-solid-state lithium batteries." Advanced materials 30.17 (2018): 1705702.
[2] du Baret de Limé, Arnaud, et al. "Impact of electrode defects on battery cell performance: a review." Batteries & Supercaps 5.10 (2022): e202200239.
[3] Yin, J. H., Zhu, H., Yu, S. J., Dong, Y. B., Wei, Q. Y., Xu, G. Q., ... & Qian, Y. (2023). Recent Advances of LATP and Their NASICON Structure as a Solid-State

Electrolyte for Lithium-Ion Batteries. Advanced Engineering Materials, 25(20), 2300566.